

Near-linear Euclidean Lipschitz selection for interval-valued maps

Automated mathematical discovery packet

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Status. Candidate partial result, likely valid, subject to human review. It fully answers the interval-valued case singled out in the source’s open Remark 1.4; the arbitrary half-plane-valued case remains open.

1 Source problem

The source is P. Shvartsman, *Efficient Algorithms for Lipschitz Selections of Set-Valued Mappings in $\mathbb{R}^2$* , arXiv:2503.19094 [1]. Its Theorem 1.3 gives a quadratic-work, quadratic-storage gap algorithm for half-plane-valued maps. On physical PDF page 4 (source TeX line 1341), Remark 1.4 asks:

“Can we replace the Projection Algorithm with an algorithm requiring at most $O(N \log N)$ operations and $O(N)$ storage, similar to the Approximate Selection Algorithm described in [35]?”

The domain in the example is an N -point subset of $\mathbb{R}^n$ with Euclidean distance. The remark then says explicitly that the answer is unclear even for $D = 1$, when every value is a closed interval in $\mathbb{R}$.

Theorem 1.3 implies an analog of Theorem 1.2 (for the case $\tau = 0$ and $D = 2$) for set-valued mappings defined on a pseudometric space and taking values in the family of all convex (not necessarily bounded) polygons in $\mathbb{R}^2$. This is Theorem 6.10 which we prove in Section 6.6.

Remark 1.4 Efficiency estimates of the Projection Algorithm given in Theorem 1.3 lead naturally to the following interesting (open) problem: Given a specific, non-trivial N -element (pseudo)metric space (M, ρ) , under what conditions can we construct a Lipschitz selection algorithm with a significantly lower computational cost than the $O(N^2)$ complexity of the Projection Algorithm presented in Theorem 1.3? For instance, let (M, ρ) be an N -point subset of $\mathbf{R}^n$ equipped with the standard Euclidean metric. *Can we replace the Projection Algorithm with an algorithm requiring at most $O(N \log N)$ operations and $O(N)$ storage, similar to the Approximate Selection Algorithm described in [35]?* (A question posed by C. Fefferman.)

We note that the answer to this question remains unclear even in the seemingly simpler one dimensional setting $D = 1$, i.e., for set-valued mappings whose images are closed intervals in $\mathbf{R}$. ◀

Figure 1: The complete open Remark 1.4, rendered from the stored arXiv source (physical PDF page 4).

2 Definitions and output convention

Let $(\mathcal{M}, \rho)$ be a finite metric space. A weighted connected graph $G = (\mathcal{M}, \mathcal{E})$ is an *upper t -spanner* for ρ if its shortest-path metric d_G satisfies

$$\rho(x, y) \leq d_G(x, y) \leq t\rho(x, y) \quad (x, y \in \mathcal{M}).$$

A usual metric spanner, whose edge xy has weight $\rho(x, y)$, has the first inequality automatically by the triangle inequality.

An interval is written $I_x = [a_x, b_x]$, allowing $a_x = -\infty$ and $b_x = +\infty$. A selection is a map f with $f(x) \in I_x$. Given $\lambda \geq 0$, the desired gap output is either

- (i) a sound certificate that no λ -Lipschitz selection exists; or
- (ii) a selection with Lipschitz constant at most $t\lambda$.

This is the same one-sided output convention as the source: the algorithm is allowed to return a controlled-factor success even when no selection exists at the smaller constant λ .

3 Proof intuition

For scalar intervals, all global compatibility information collapses to one min-plus distance transform. With the complete metric it is

$$x \mapsto \min_y \{b_y + \lambda\rho(x, y)\}.$$

Replacing ρ by an upper spanner metric can only increase this transform. Consequently, failure for the spanner is still a genuine failure for the original metric. On the other hand, success is Lipschitz for the spanner metric and loses only its stretch factor when returned to ρ .

The transform is exactly a multi-source shortest-path computation. A fixed-dimensional Euclidean $(1 + \varepsilon)$ -spanner has only linearly many edges and is constructible in near-linear time, so one shortest-path pass delivers the requested complexity. The remaining $\mathbb{R}^2$ half-plane problem does not collapse to a single scalar transform because its constraints have varying normal directions.

4 Spanner reduction

Lemma 1 (exact interval criterion). *Suppose all $I_x = [a_x, b_x]$ are nonempty. There is a λ -Lipschitz selection for $(\mathcal{M}, \rho)$ if and only if*

$$a_x \leq b_y + \lambda\rho(x, y) \quad (x, y \in \mathcal{M}).$$

Proof. Necessity follows from $a_x \leq f(x) \leq f(y) + \lambda\rho(x, y) \leq b_y + \lambda\rho(x, y)$.

If at least one b_y is finite, put

$$u(x) = \min_{y: b_y < +\infty} \{b_y + \lambda\rho(x, y)\}.$$

The assumed inequalities give $a_x \leq u(x)$, while the term $y = x$ gives $u(x) \leq b_x$ whenever b_x is finite. Thus $u(x) \in I_x$. The triangle inequality gives $u(x) \leq u(z) + \lambda\rho(x, z)$ and its symmetric version, so u is λ -Lipschitz. If every $b_y = +\infty$, the constant equal to the largest finite lower endpoint is a selection; use zero if all lower endpoints are $-\infty$. $\square$

Theorem 1 (sparse-spanner interval algorithm). *Let $G = (\mathcal{M}, \mathcal{E})$ be an upper t -spanner for a metric ρ , let $I_x = [a_x, b_x]$ be nonempty closed intervals, and let $\lambda \geq 0$. There is an algorithm with work*

$$O((|\mathcal{M}| + |\mathcal{E}|) \log |\mathcal{M}|)$$

and storage $O(|\mathcal{M}| + |\mathcal{E}|)$ which produces one of the following:

- (i) *a correct conclusion that no λ -Lipschitz selection exists for $(\mathcal{M}, \rho)$; or*
- (ii) *a selection f with $\|f\|_{\text{Lip}(\mathcal{M}, \rho)} \leq t\lambda$.*

Proof. If every upper endpoint is infinite, use the constant selection from Lemma 1. Otherwise define

$$u_G(x) = \min_{y: b_y < +\infty} \{b_y + \lambda d_G(x, y)\}.$$

If $a_x > u_G(x)$ for some x , choose a minimizer y . Since $d_G(x, y) \geq \rho(x, y)$,

$$a_x > b_y + \lambda d_G(x, y) \geq b_y + \lambda \rho(x, y).$$

This violates the necessary condition in Lemma 1, so the no-go conclusion is sound.

Otherwise $u_G(x) \in I_x$: the lower inequality is the passed test, and the upper inequality follows from the term $y = x$ whenever b_x is finite. As in the proof of the lemma,

$$|u_G(x) - u_G(z)| \leq \lambda d_G(x, z) \leq t\lambda \rho(x, z).$$

Thus u_G is the promised selection.

For the complexity claim, let B be the smallest finite upper endpoint. Initialize each vertex y with finite upper endpoint at label $b_y - B \geq 0$ and give every graph edge e length $\lambda w(e) \geq 0$. A single multi-source version of Dijkstra's label-setting algorithm [3] computes $u_G - B$. Its work and storage have the stated bounds. $\square$

5 Near-linear Euclidean consequence

Corollary 1 (the source's interval case). *Fix $n \in \mathbb{N}$ and $\varepsilon > 0$. Given an N -point set $\mathcal{M} \subset \mathbb{R}^n$, nonempty closed intervals I_x , and $\lambda \geq 0$, one can produce either*

- (i) *a sound certificate that no λ -Lipschitz interval selection exists; or*
- (ii) *an interval selection with Euclidean Lipschitz constant at most $(1 + \varepsilon)\lambda$,*

using $O_{n,\varepsilon}(N \log N)$ operations and $O_{n,\varepsilon}(N)$ storage.

Proof. Standard fixed-dimensional Euclidean spanner constructions produce a $(1 + \varepsilon)$ -spanner with $O_{n,\varepsilon}(N)$ edges in $O_{n,\varepsilon}(N \log N)$ time and linear storage [2]. Apply Theorem 1 with $t = 1 + \varepsilon$. $\square$

This gives a full affirmative answer to the $D = 1$ case that Remark 1.4 explicitly singles out. When the domain lies in $\mathbb{R}$, sorting the points and joining consecutive points produces an exact 1-spanner, so the success constant improves to λ itself.

The same proof coordinate by coordinate gives another extension: for fixed D , if every value is an axis-aligned box in $(\mathbb{R}^D, \|\cdot\|_\infty)$, the same gap algorithm uses $O_{n,D,\varepsilon}(N \log N)$ work and linear storage and returns a $(1 + \varepsilon)\lambda$ -selection. A no-go in one coordinate is a valid no-go for the box problem.

6 Verification and upgrade attempts

The script `code/verify_spanner_interval_selection.py` checks 600 random line instances using the exact path spanner and 600 random planar instances using small greedy 1.8-spanners. It includes rays and the whole line. In every case it verifies the spanner inequalities, compares no-go outputs with the complete-metric criterion of Lemma 1, and checks interval membership and the promised Lipschitz bound. It prints **VERDICT: PASS**. The experiment is a regression check, not the proof.

Eight focused upgrade routes are recorded in the run’s **attempts/** directory. The first four establish the exact scalar criterion, the abstract spanner transfer, the Euclidean near-linear bound, and the unbounded/box extensions. The remaining routes test an exact factor-one distance transform, a sparse half-plane feasibility reduction, WSPD compression of the source algorithm, and approximate convex optimization. They do not yield the full arbitrary-half-plane algorithm.

7 Literature and novelty check

The bounded search covered the exact title and arXiv id, the exact phrase “remains unclear even in the seemingly simpler one dimensional setting”, the author together with interval/Lipschitz-selection/near-linear keywords, and combinations with Euclidean spanners and shortest paths. Searches used arXiv and primary journal or conference pages. No later paper stating this interval subproblem or the spanner solution was found.

Novelty confidence is moderate-to-high. The two ingredients are elementary once juxtaposed, and the Euclidean spanner theorem is classical, so an unindexed prior observation remains possible. However, the 2025 source itself explicitly labels the interval case unclear, and the bounded 2026 search found no subsequent answer.

8 Limitations and human-review recommendation

The near-linear constants assume fixed ambient dimension n and fixed ε ; the claim is not a dimension-uniform algorithm. The result fully resolves only interval-valued maps and, by products, axis-aligned boxes. It does not solve the source’s arbitrary half-plane or polygon problem in $\mathbb{R}^2$.

For that remaining case, spanner edges reduce the problem to a sparse linear feasibility system, but no exact near-linear solver or compressed version of the Projection Algorithm is supplied. Human review should check the direction of the two spanner inequalities in the no-go and success arguments and confirm that the cited Euclidean spanner construction matches the source’s fixed-dimensional real-RAM model. All other steps are elementary.

References

- [1] P. Shvartsman, *Efficient Algorithms for Lipschitz Selections of Set-Valued Mappings in $\mathbf{R}^2$* , arXiv:2503.19094; J. Approx. Theory 316 (2026), article 106277, doi:10.1016/j.jat.2025.106277.
- [2] S. Arya, G. Das, D. M. Mount, J. S. Salowe, and M. Smid, *Euclidean spanners: short, thin, and lanky*, Proc. 27th ACM Symposium on Theory of Computing (1995), 489–498, doi:10.1145/225058.225191.
- [3] E. W. Dijkstra, *A note on two problems in connexion with graphs*, Numer. Math. 1 (1959), 269–271, doi:10.1007/BF01386390.